

Spike-train statistics

This document covers Poisson processes, the Fano factor and coefficient of variation, and spike-triggered averages. You will explore these topics (and others) in the extended coursework, and may see them on the exam.

These notes are a shorter form of Dayan and Abbott sections 1.4, 2.2, and 2.4. Students are encouraged to look through these sections, which provide a more lengthy coverage that some may find clearer.

Poisson processes

A **Poisson process** is a memoryless random process producing a time series of events; In the application to spike trains an event corresponds to a spike. **Memorylessness** means that the chance of an event depends only on a rate¹ function $\lambda(t)$ and not on the history of past events. The probability of an event in a small interval Δt wide is $\lambda(t) \cdot \Delta t$.

Imagine a person fishing in the sea. The chance of catching a fish might depend on the time of day but it doesn't depend on how many fish the person has already caught. However, if they are fishing in a small pond the chance of catching a fish would diminish as they catch fish. Fishing in the sea is a Poisson process; fishing in a pond is not.

We will begin by exploring a **homogeneous** Poisson process, where here *homogeneous* means that its rate λ is constant. Of course, if there is a refractory period (a time during which the neuron cannot spike), the rate is not constant in time, and the spike train is not a homogeneous Poisson process. In neuroscience, it is common to treat spike trains as Poisson processes with time-varying "*inhomogeneous*" rates. This is called an **inhomogeneous Poisson process**.

For a homogeneous Poisson process, one can prove—with a nice argument you are urged to look up—that the distribution of inter-event intervals t , here inter-spike intervals, is an exponential distribution given by

$$p(t) = \lambda e^{-\lambda t} \tag{1}$$

¹some sources may use the term "intensity" rather than "rate"

Aside: In the literature, you may also see the Poisson process process called a *Poisson point process* or simply *point process*. Neuroscience also uses point processes that are not technically Poisson processes. These appear when we model, for example, the refractory period as a change in $\lambda(t)$ that depends on past spiking outputs. This violates the memorylessness property (refractoriness is like the “fishing in a small pond” metaphor, above). For example, Autoregressive Generalized Linear Point-Process Models (PP-GLMs) are a type of point-process model that treats a nonlinear function of the intensity λ as a linear function of other time-series variables, and the neuron’s own spiking history. These models are used in correlative analyses to create statistical tests that measure which quantities co-vary with a neuron’s spiking and how.

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For a homogeneous Poisson process, the Fano factor “ F ” of binned spike counts, and the coefficient of variation C_v of the inter-spike-interval distribution, are both 1.

These quantities measure how regular a neuron’s spiking is. Neurons that have a Fano factor larger than one may have additional processes that cause their firing rate to vary over time. Neurons that have a Fano factor less than one are more regular than expected for a homogeneous Poisson process. We will calculate both Fano factor and coefficient of variation for simulated and real spike trains.

Fano factor: To calculate the Fano factor, we count the number of spikes k occurring in bins over a longer interval of time Δt_{bin} . The Fano factor is then defined as

$$F = \frac{\sigma_k^2}{\mu_k} \quad (2)$$

where σ_k^2 is the variance of binned spike counts K , and μ_k the average.

Coefficient of variation: For spike trains, we usually apply the coefficient of variation to the *Inter-Spike Interval* (or ISI for short) distribution. This is the distribution of times between consecutive spikes. The coefficient of variation is defined as:

$$C_v = \frac{\sigma_{\text{ISI}}}{\mu_{\text{ISI}}} \quad (3)$$

where σ_{ISI} is a standard deviation of the time between consecutive spikes, and μ_{ISI} is an average time between spikes.

The spike triggered average (STA) The Spike Triggered Average (STA) is used to identify what is causing the neuron to spike. Imagine the neuron spikes at times $\{t_1, t_2, \dots, t_n\}$ and the stimulus is $x(t)$ then the spike triggered average is

$$s(\tau) = \frac{1}{n} \sum_{i=1}^n x(t_i - \tau) \quad (4)$$

In other words, it is the average value of stimulus a time offset τ relative to when the neuron spikes.